
Pierian Data Inc. Presents

Data Science Career Interview Prep Guide

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Table of Contents

[Table of Contents](#)

[Course Overview](#)

[Data Science Career Overview](#)

[DS Career Overview Slides:](#)

[Why Choose a Career in Data Science?](#)

[Data Science is Interdisciplinary](#)

[Data Science Positions and Titles](#)

[Data Science Interview Preparation](#)

[Technical Tools of the Trade](#)

[Theory Knowledge](#)

[Machine Learning Knowledge](#)

[Software Knowledge](#)

[Data Science Interview Process](#)

[Resumes](#)

[Interview Process](#)

[Landing Interviews](#)

[Negotiating Offers](#)

[Probability Theory Resources](#)

[Probability Slides:](#)

[Probability Question 1 Help - Consecutive Coin Flip](#)

[Probability Question 2 Help - Dice roll sum of 4 Odds](#)

[Probability Question 3 Help - Dice roll with at least one 4](#)

[Probability Question 4 Help - Red and Blue Marbles](#)

[Probability Question 5 Help - Odds of Car in 10 Minutes](#)

[Probability Question 6 Help - Average Flips for 2 Heads in a Row](#)

[Probability Question 7 Help - 1 Biased Coin of 10](#)

[Probability Question 8 Help - Simulating Fair Coin from Biased Coin](#)

[Probability Question 9 Help - Alice, Odds of having another Girl](#)

[Statistics Resources](#)

[Statistic Slides:](#)

[Statistics Question 1 Help - Raining in Seattle](#)

[Statistics Question 2 Help - Quantum Messaging](#)

[Statistics Question 3 Help - Type I vs Type II Errors](#)



[Statistics Question 4 Help - New Virus Test](#)

[Statistics Question 5 Help - Motor Life Guarantee](#)

[Product Design and Metrics Resources](#)

[Product Design and Metrics Slides:](#)

[Product Design and Metrics Interview Question 1 - FB Messenger](#)

[Product Design and Metrics Interview Question 2,3,4 - AIB Google](#)

[Product Design and Metrics Interview Question 5 - Car Gas Usage](#)

[Working with Data with SQL Resources](#)

[SQL Interview Question 1 - Query Check](#)

[SQL Interview Question 2 - Query Check](#)

[SQL Interview Question 3 - Query Check](#)

[SQL Interview Question 4 - Employees and Managers](#)

[SQL Interview Question 5 - Employees and Managers](#)

[Machine Learning Resources](#)

[Machine Learning Slides](#)

[Machine Learning Question 1 - Linear Regression Assumptions](#)

[Machine Learning Question 2 - Logistic Regression](#)

[Machine Learning Question 3 - Decision Tree Splits](#)

[Machine Learning Question 4 - Decision Tree Advantages](#)

[Machine Learning Question 5 - Random Forest vs Boosting](#)

[Machine Learning Question 6 - Naive Bayes Assumptions](#)

[Machine Learning Question 7 - How SVM Works](#)

[Machine Learning Question 8 - Overfitting](#)

[Machine Learning Question 9 - Accuracy, Precision, Recall](#)

[Machine Learning Question 10 - Regression Metrics](#)

[Design of Experiments Resources](#)

[Design of Experiments Slides:](#)

[Design of Experiments Question 1 - A/B Testing](#)

[Design of Experiments Question 2 - Biases](#)

[Design of Experiments Question 3 - Multiple Versions of Page Testing](#)

[Design of Experiments Question 4](#)

[Coding Resources](#)



Course Overview

Welcome to the course!

This guide contains the resources and links mentioned in the lecture in one convenient location.

If during the course videos you hear us mention resources or links, just open up this guidebook to find them!

You can use the Table of Contents as links to directly jump to whatever location you need to go to.

Best of luck on your data science journey!

If you ever have a need for corporate training, feel free to reach us at training@pieriandata.com.

Data Science Career Overview

DS Career Overview Slides:

<https://docs.google.com/presentation/d/1V1GDbnU00fZbejq3U61e3D-TqB0FxyrlyURpDtAEH8/edit?usp=sharing>

Why Choose a Career in Data Science?

Deciding on a Data Science Career:

<https://www.kdnuggets.com/2014/03/data-scientist-right-career-path-candid-advice.html>

<http://www.datacenterjournal.com/consider-career-data-science/>

Data Science is Interdisciplinary

Overviews of Data Science:

<https://datascience.nyu.edu/what-is-data-science/>

https://en.wikipedia.org/wiki/Data_science

Data Science Positions and Titles

Large list of various job titles:

<https://www.datasciencecentral.com/profiles/blogs/job-titles-for-data-scientists>

<https://www.datasciencecentral.com/profiles/blogs/400-categorized-job-titles-for-data-scientists>

Thoughts on Higher Education

Other opinions on higher education:

<https://www.kdnuggets.com/2014/06/masters-degree-become-data-scientist.html>

Data Science Interview Preparation

Technical Tools of the Trade

Python:

<https://www.python.org/>

R:

<https://www.r-project.org/about.html>

Theory Knowledge

Free Online Stats Books:

<http://onlinestatbook.com/>

<https://www.openintro.org/stat/textbook.php>

Machine Learning Knowledge

Great Book on Machine Learning (Free):

<http://www-bcf.usc.edu/~gareth/ISL/ISLR%20Seventh%20Printing.pdf>

Software Knowledge

Tableau:

<https://www.tableau.com/>



Data Science Interview Process

Resumes

Example Data Science Resume:

<http://will-stanton.com/creating-a-great-data-science-resume/>

https://docs.google.com/document/d/1ktqmnyG8oInZ24DqJZ3G9h619ng-JYeiY6gNh-i_Q6o/edit

<https://www.monster.com/career-advice/article/data-scientist-resume-sample>

Interview Process

AirBnb's Interview Process:

<https://medium.com/@AirbnbCandidateJourney/leveling-the-playing-field-an-overview-of-airbnb-s-data-science-interview-process-bd0660b77a17>

Nice Guide to Interviews:

<https://www.linkedin.com/pulse/how-ace-data-science-interview-vin-vashishta/>

Someone's experience with DS Interviews

<https://alyaabbott.wordpress.com/2014/10/01/how-to-ace-a-data-science-interview/>

Nice write ups on other individual's process:

<https://medium.com/@XiaohanZeng/i-interviewed-at-five-top-companies-in-silicon-valley-in-five-days-and-luckily-got-five-job-offers-25178cf74e0f>

More thoughts on interviews:

<https://www.quora.com/How-do-I-prepare-for-a-data-scientist-interview>



Landing Interviews

Experience Essay Blog Post:

<http://www.erinshellman.com/crushed-it-landing-a-data-science-job/>

General Discussion:

http://treycausey.com/data_science_interviews.html

Nide Guide:

<http://data-informed.com/six-secrets-to-landing-a-job-in-data-science/>

Negotiating Offers

Guides for Salary Negotiation:

<https://www.nerdwallet.com/blog/loans/student-loans/negotiate-salary-evaluate-offer/>

<https://www.themuse.com/advice/how-to-negotiate-salary-37-tips-you-need-to-know>

<https://www.thebalance.com/salary-negotiation-tips-how-to-get-a-better-offer-2063439>

<https://www.washingtonpost.com/graphics/business/womens-wages/salary-negotiation-guide-women/>

Probability Theory Resources

Probability Slides:

<https://docs.google.com/presentation/d/13M9iCeBU8fngtU4tzjudMCmmP6Fd3BsCUuycCuH3PIE/edit?usp=sharing>

Probability Question 1 Help - Consecutive Coin Flip

<https://math.stackexchange.com/questions/112726/coin-tossed-until-two-consecutive-heads-or-tails-appear>

https://en.wikipedia.org/wiki/Geometric_series#Sum

https://en.wikipedia.org/wiki/Expected_value#Finite_case

Probability Question 2 Help - Dice roll sum of 4 Odds

<http://alumnus.caltech.edu/~leif/FRP/probability.html>

Probability Question 3 Help - Dice roll with at least one 4

<http://alumnus.caltech.edu/~leif/FRP/probability.html>

Probability Question 4 Help - Red and Blue Marbles

<https://www.techinterview.org/post/526363745/red-marbles-blue-marbles/>

Probability Question 5 Help - Odds of Car in 10 Minutes

<https://www.quora.com/If-the-probability-of-observing-a-car-in-30-minutes-on-a-highway-is-0-95-what-is-the-probability-of-observing-a-car-in-10-minutes-assuming-constant-default-probability>

<https://math.stackexchange.com/questions/52113/probability-calculations-on-highway>

Probability Question 6 Help - Average Flips for 2 Heads in a Row

<https://www.codechef.com/wiki/tutorial-expectation>

https://courses.cit.cornell.edu/info2950_2012sp/mh.pdf



Probability Question 7 Help - 1 Biased Coin of 10

https://en.wikipedia.org/wiki/Bayes%27_theorem

Probability Question 8 Help - Simulating Fair Coin from Biased Coin

<http://www.eecs.harvard.edu/~michaelm/coinflipext.pdf>

<https://jeremykun.com/2014/02/08/simulating-a-fair-coin-with-a-biased-coin/>

Probability Question 9 Help - Alice, Odds of having another Girl

<https://math.stackexchange.com/questions/15055/in-a-family-with-two-children-what-are-the-chances-if-one-of-the-children-is-a>



Statistics Resources

Statistic Slides:

https://docs.google.com/presentation/d/1VCLkMu3CRTuSN-nC4EaPVwhXd9a_0p-4S-ci5FJRJKI/edit?usp=sharing

Statistics Question 1 Help - Raining in Seattle

<https://www.mathsisfun.com/data/probability-tree-diagrams.html>

Statistics Question 2 Help - Quantum Messaging

<https://www.intmath.com/counting-probability/12-binomial-probability-distributions.php>

Statistics Question 3 Help - Type I vs Type II Errors

https://en.wikipedia.org/wiki/Type_I_and_type_II_errors

Statistics Question 4 Help - New Virus Test

<https://math.hmc.edu/funfacts/tag/probability/>

Statistics Question 5 Help - Motor Life Guarantee

<http://www.z-table.com/>

<https://statistics.laerd.com/statistical-guides/normal-distribution-calculations.php>

Product Design and Metrics Resources

Product Design and Metrics Slides:

<https://docs.google.com/presentation/d/1pY15hTUjMyzDEl-RenE-hvwJsfZNIJCp4TRVkJGsLR0o/edit?usp=sharing>

Product Design and Metrics Interview Question 1 - FB Messenger

<https://savvyapps.com/blog/mobile-app-analytics>

Product Design and Metrics Interview Question 2,3,4 - A/B Google

https://en.wikipedia.org/wiki/A/B_testing

<https://www.pardot.com/blog/abcs-ab-testing/>

<http://online-behavior.com/testing/advanced-ab-testing-tactics-1356>

Product Design and Metrics Interview Question 5 - Car Gas Usage

https://www.glassdoor.com/Interview/business-sense-There-are-two-types-of-cars-A-and-B-The-number-of-people-in-US-who-use-A-and-B-are-the-same-They-drive-QTN_831574.htm



Working with Data with SQL Resources

SLIDES:

https://docs.google.com/presentation/d/1YXmZ1Uu9TL5ckxNOk8c7k8a6dd3Lk_DbG30hOPatQVg/edit?usp=sharing

SQL Interview Question 1 - Query Check

https://www.w3schools.com/sql/sql_alias.asp

SQL Interview Question 2 - Query Check

https://www.w3schools.com/sql/sql_groupby.asp

SQL Interview Question 3 - Query Check

https://www.w3schools.com/sql/sql_having.asp

SQL Interview Question 4 - Employees and Managers

<https://www.w3schools.com/sql/default.asp>

SQL Interview Question 5 - Employees and Managers

https://www.w3schools.com/sql/sql_join.asp

Machine Learning Resources

Machine Learning Slides

https://docs.google.com/presentation/d/1VvSWus6sjXEV7WG7FuH__Pi5rwB0_WIKc7A5Whx5gVg/edit?usp=sharing

Machine Learning Question 1 - Linear Regression Assumptions

<http://www.statisticssolutions.com/assumptions-of-linear-regression/>

<http://r-statistics.co/Assumptions-of-Linear-Regression.html>

<https://stats.stackexchange.com/questions/16381/what-is-a-complete-list-of-the-usual-assumptions-for-linear-regression>

Machine Learning Question 2 - Logistic Regression

https://en.wikipedia.org/wiki/Logistic_regression

<http://dataaspirant.com/2017/03/02/how-logistic-regression-model-works/>

Machine Learning Question 3 - Decision Tree Splits

https://en.wikipedia.org/wiki/Information_gain_in_decision_trees

<http://dni-institute.in/blogs/cart-decision-tree-gini-index-explained/>

<https://link.springer.com/article/10.1023/B:AMAI.0000018580.96245.c6>

Machine Learning Question 4 - Decision Tree Advantages

https://en.wikipedia.org/wiki/Decision_tree_learning

<http://www.brighthubpm.com/project-planning/106000-advantages-of-decision-tree-analysis/>

<http://www.simafore.com/blog/bid/62333/4-key-advantages-of-using-decision-trees-for-predictive-analytics>

Machine Learning Question 5 - Random Forest vs Boosting

<http://fastml.com/what-is-better-gradient-boosted-trees-or-random-forest/>

<https://www.quora.com/When-would-one-use-Random-Forests-over-Gradient-Boosted-Machines-GBMs>

<https://discuss.analyticsvidhya.com/t/what-is-the-fundamental-difference-between-randomforest-and-gradient-boosting-algorithms/2341>

<https://stats.stackexchange.com/questions/173390/gradient-boosting-tree-vs-random-forest>

Machine Learning Question 6 - Naive Bayes Assumptions

<https://nlp.stanford.edu/IR-book/html/htmledition/properties-of-naive-bayes-1.html>

https://en.wikipedia.org/wiki/Naive_Bayes_classifier

Machine Learning Question 7 - How SVM Works

https://en.wikipedia.org/wiki/Support_vector_machine

Machine Learning Question 8 - Overfitting

<https://en.wikipedia.org/wiki/Overfitting>

<https://www.quora.com/What-is-an-intuitive-explanation-of-over-fitting-particularly-with-a-small-sample-set-What-are-you-essentially-doing-by-over-fitting-How-does-the-over-promise-of-a-high-R%C2%B2-low-standard-error-occur>

Machine Learning Question 9 - Accuracy, Precision, Recall

https://en.wikipedia.org/wiki/Precision_and_recall

Machine Learning Question 10 - Regression Metrics

<https://people.duke.edu/~rnau/compare.htm>

Design of Experiments Resources

Design of Experiments Slides:

<https://docs.google.com/presentation/d/1mmfJyhApROEfTZrNAIFIGJfS0IRDsJXD3Fr2WhhgxEI/edit?usp=sharing>

Design of Experiments Question 1 - A/B Testing

<https://www.optimizely.com/optimization-glossary/ab-testing/>

https://en.wikipedia.org/wiki/A/B_testing

Design of Experiments Question 2 - Biases

[https://en.wikipedia.org/wiki/Bias_\(statistics\)](https://en.wikipedia.org/wiki/Bias_(statistics))

<http://www.statisticshowto.com/what-is-bias/#Availability>

Design of Experiments Question 3 - Multiple Versions of Page Testing

https://en.wikipedia.org/wiki/Multi-armed_bandit

<https://www.searchenginepeople.com/blog/16072-multi-armed-bandits-ab-testing-makes-money.html>

Design of Experiments Question 4

<http://rpsychologist.com/d3/NHST/>

<https://onlinecourses.science.psu.edu/stat414/book/export/html/245>

https://en.wikipedia.org/wiki/Statistical_power



Coding Resources

Coding Slides

<https://docs.google.com/presentation/d/15SCi6Fpiq0edRYhM5w7GDZBm3kdeUsWg8kSEHVziYZ/g/edit?usp=sharing>

Coding Interview Question 1 - Largest Continuous Sum

Solution Link in Python:

<https://gist.github.com/Pierian-Data/51202d7a84e36537df97a7c9ca78061f>

Coding Interview Question 2 - String Compression

Solution Link in Python:

<https://gist.github.com/Pierian-Data/4c85a5adc36a282223065d52cc178595>

Coding Interview Question 3 - Stock Prices

Solution Link in Python:

<https://gist.github.com/Pierian-Data/a17fd7357aeb919306864fe435e6ed15>

Coding Interview Question 4 - Missing Element

Solution Link in Python:

<https://gist.github.com/Pierian-Data/ea5589bb6ad88a5ec00b42d8d0f29328>